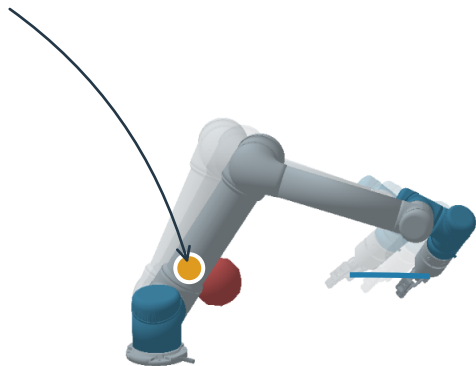


(a) Hold the same joint velocity

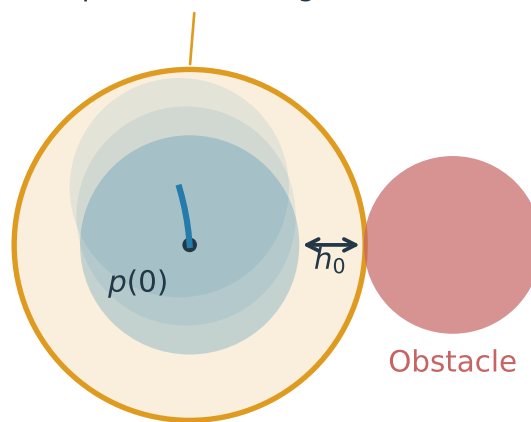
Track a collision sphere



Solid: initial pose • Ghosts: held control

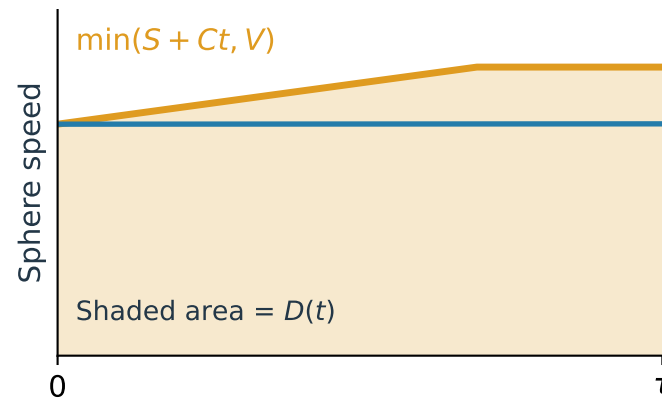
(b) Bound the whole sweep

Computed enclosing ball



Every moved sphere lies inside $B(p(0), r + D(t))$

(c) Integrate the speed bound



Held time
 $\|p(t) - p(0)\| \leq D(t) \leq h_0$

The swept sphere cannot reach the obstacle.

Computed UR5 example: certified hold 0.421 s | Initial clearance 50.3 mm